

Use Case Diagram

Aim: Draw Use Case Diagram and write use case description for the project.

Use Case Descriptions-B&B Reservation System

Use Case 1: Register

Summary:

Allows a customer to create a new account.

Actors:

Customer

Pre-conditions:

- Customer is not registered

Description:

1. Customer selects register option
2. System displays form
3. Customer enters details
4. System validates data
5. Account created successfully

Exceptions:

- Invalid input

Post-conditions:

- Customer account created

Use Case 2: Login

Summary:

Allows registered users to access the system.

Actors:

Customer, Owner

Pre-conditions:

- User registered

Description:

1. User enters username and password
2. System verifies credentials
3. Dashboard displayed

Exceptions:

- Invalid credentials

Post-conditions:

- User logged in

Use Case 3: Check Room Availability

Summary:

Allows the customer to check available rooms for selected dates before making a reservation.

Actors:

Customer

Pre-conditions:

- User logged in

Description:

1. Customer enters check-in and check-out dates
2. System checks database
3. Available rooms displayed

Exceptions:

- No rooms available

Post-conditions:

- Available rooms list shown

Use Case 4: Reserve Room**Summary:**

Allows the customer to book a selected room for chosen dates.

Actors:

Customer

Pre-conditions:

- Room available

Description:

1. Customer selects room
2. Enters reservation details
3. System saves reservation

Exceptions:

- Room already booked

Post-conditions:

- Reservation stored

Use Case 5: Make Payment**Summary:**

Allows the customer to complete advance payment to confirm reservation.

Actors:

Customer

Pre-conditions:

- Reservation created

Description:

1. Customer enters payment details
2. System validates payment
3. Reservation confirmed

Exceptions:

- Payment failed

Post-conditions:

- Payment recorded

Use Case 6: View Reservations**Summary:**

Allows the owner to see all customer reservations.

Actors:

Owner

Pre-conditions:

- Owner logged in

Exceptions:

- No reservations found
- System unable to retrieve reservation data

Description:

1. Owner selects view reservations
2. System displays reservation list

Post-conditions:

- Reservations visible

Use Case 7: Manage Rooms

Summary:

Allows the owner to add, update, or change room status.

Actors:

Owner

Pre-conditions:

- Owner must be logged in
- Room data must be available in system

Exceptions:

- Invalid room details entered
- System error while updating room data

Description:

1. Owner updates room status
2. System saves changes

Post-conditions:

- Room information is successfully updated
- Updated room status is reflected in system

Use Case 8: Add Expense

Summary:

Allows the owner to record business expenses.

Actors:

Owner

Pre-conditions:

- Owner must be logged in

Exceptions:

- Missing expense information
- Invalid amount entered

Description:

1. Owner enters expense details
2. System stores data

Post-conditions:

- Expense details are stored in database

Use Case 9: View Profit Report**Summary:**

Allows the owner to view income, expenses, and net profit.

Actors:

Owner

Pre-conditions:

- Owner must be logged in
- Income and expense records must exist

Description:

1. Owner selects report period
2. System calculates totals
3. Report displayed

Exceptions:

- No data available for selected date range
- System calculation error

Post-conditions:

- Profit report is generated successfully
- Income and expense totals are displayed

Use Case 10: Logout

Summary:

Allows users to securely exit the system.

Actors:

Customer, Owner

Pre-conditions:

- User must be logged in

Description:

1. User clicks logout
2. System ends session
3. Login page displayed

Exceptions:

- Session already expired
- System timeout

Post-conditions:

- User session is terminated
- User is redirected to login page